Chapter 6, Problem 1P

Problem

If the voltage across a 7.5-F capacitor is 2*te*−3*t* V, find the current and the power.

Step-by-step solution

Step 1 of 1

The value of capacitance, $C = 7.5\ \text{F}$

The voltage across capacitor, $v(t) = 2te^{-3t}\ \text{V}$

Determine the current through the capacitor.

$$\begin{aligned} i(t) &= C\frac{dv(t)}{dt} \\ &= (7.5)\frac{d}{dt}\left(2te^{-3t}\right) \\ &= (7.5\times 2)\left[e^{-3t} + t(-3)e^{-3t}\right] \\ &= 15\left[e^{-3t}(1-3t)\right]\ \text{A} \end{aligned}$$

Thus, the current through the capacitor is $\boxed{15\left[(1-3t)e^{-3t}\right]\ \text{A}}$.

Determine the power in the capacitor using the following formula:

$$p(t) = v(t)\times i(t)$$

$$\begin{aligned} p(t) &= 15\left[e^{-3t}(1-3t)\right]\left(2te^{-3t}\right) \\ &= 30t(1-3t)e^{-6t}\ \text{W} \end{aligned}$$

Thus, the power in the capacitor is $\boxed{30t(1-3t)e^{-6t}\ \text{W}}$.